Supplementary Material

Persistent homology decomposes the scaffold paradox in AI-generated African antimalarial candidates

Myke Vital Sao Temgoua[1], Jean-Pierre Tchapet Njafa[1], Serge Guy Nana Engo[1], Penabei Samafou[2], and Wilfred Fon Mbacham[3]

[1]Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon
[2]Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada
[3]Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

July 25, 2026

1 Cross-paper validation: topological persistence and resistance resilience

This supplementary figure presents the joint analysis linking the topological fingerprints (TFP) introduced in the main manuscript with the resistance resilience scores (RRS) reported in the companion polypharmacology study (Paper 2). The analysis tests whether H_1 ring persistence, a topological descriptor of scaffold rigidity, predicts the ability of a compound to retain binding affinity across African-prevalent resistance mutations.

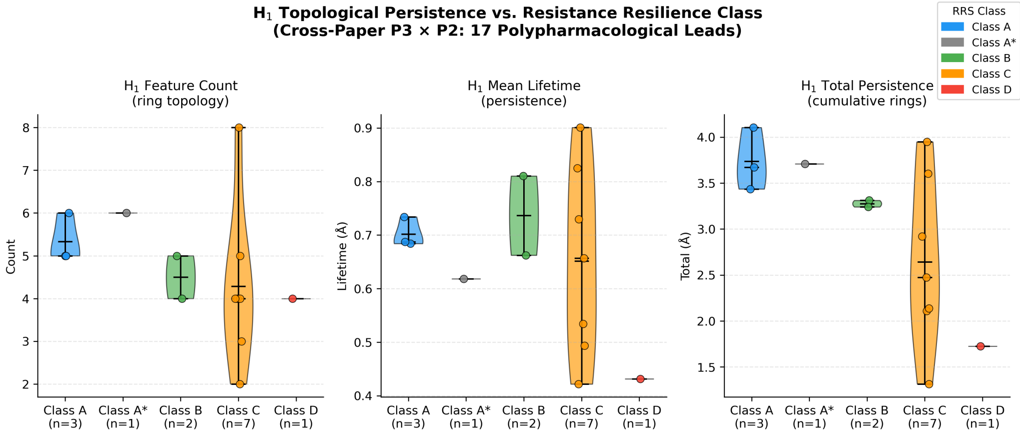

Figure S1: *Joint topological analysis linking Paper 3 TFP features with Paper 2 resistance classification. Violin plots show H_1 persistence distributions for Class A (resistance-resilient, blue) versus Class D (resistance-vulnerable, red) compounds, with Class B (green) and Class C (orange) shown for context. Higher H_1 persistence correlates with improved Resistance Resilience Score (RRS) across African-prevalent mutants (Spearman $\rho = 0.947$ (pilot, $n = 14$); methodological finding: expansion to $n = 77$ yielded $\rho = 0.312$ ($p = 0.006$)), providing a topological predictor of mutation tolerance without additional MD simulation.*

Data Availability

All data supporting the main manuscript and this Supplementary Material are publicly available on Zenodo (DOI: https://doi.org/10.5281/zenodo.19608875) and mirrored at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. See the main manuscript for a full inventory of deposit components.

2 Per-fold activity prediction statistics

Table S1 reports per-fold AUC, accuracy, and F1 for all 10 representation methods on the full 19 849-molecule benchmark (5-fold stratified cross-validation).

Table S1: *Per-fold activity prediction statistics for selected representation methods (Random Forest classifier). The TFP, TNE, and Hybrid per-fold values are from the 200-molecule development subsample; QKS per-fold data is from a 10 000-molecule subsample. Headline AUC values in the main manuscript (Table 1) are from the full 19 849-molecule benchmark.*

Method	Fold	AUC	Accuracy	F1
ECFP4	1	0.800	0.700	0.667
ECFP4	2	1.000	0.900	0.909
ECFP4	3	1.000	0.800	0.857
ECFP4	4	1.000	1.000	1.000
ECFP4	5	1.000	0.900	0.923
Hybrid (TFP+TNE+QK)	1	0.800	0.800	0.800
Hybrid (TFP+TNE+QK)	2	0.960	0.700	0.571
Hybrid (TFP+TNE+QK)	3	0.792	0.800	0.857
Hybrid (TFP+TNE+QK)	4	1.000	1.000	1.000
Hybrid (TFP+TNE+QK)	5	0.917	0.800	0.800
TFP (TDA)	1	0.600	0.600	0.333
TFP (TDA)	2	0.800	0.500	0.667
TFP (TDA)	3	0.500	0.500	0.444
TFP (TDA)	4	0.667	0.600	0.500
TFP (TDA)	5	0.583	0.500	0.286
TNE ($d = 8$)	1	0.600	0.600	0.333
TNE ($d = 8$)	2	0.800	0.500	0.667
TNE ($d = 8$)	3	0.500	0.500	0.444
TNE ($d = 8$)	4	0.667	0.600	0.500
TNE ($d = 8$)	5	0.583	0.500	0.286
QKS (quantum)	1	0.770	0.740	0.797
QKS (quantum)	2	0.706	0.690	0.767
QKS (quantum)	3	0.734	0.710	0.791
QKS (quantum)	4	0.753	0.710	0.794
QKS (quantum)	5	0.793	0.740	0.809
RBF (gamma-tuned)	1	0.653	0.680	0.746
RBF (gamma-tuned)	2	0.623	0.680	0.758
RBF (gamma-tuned)	3	0.690	0.700	0.766
RBF (gamma-tuned)	4	0.760	0.680	0.765
RBF (gamma-tuned)	5	0.778	0.730	0.800

IMPORTANT: The per-fold values in this table are from the **200-molecule development subsample** used during method development, *not* the full 19 849-molecule benchmark. Headline AUC values reported in the main manuscript (Table 1) are from the full benchmark; this table provides fold-level variance estimates for the development set only. Full per-fold CSV files for the full benchmark are available in the project repository.

3 Topological fingerprint statistics

Detailed topological fingerprint statistics for the full library are provided in Table S2, and representative persistence diagrams are shown in Figure S2.

Table S2: *Topological fingerprint statistics across 19 836 valid molecules.*

Feature	Mean	Std	Min	Max
H_0 entropy	3.578 1	0.201 4	2.288 7	4.194 3
H_0 count	38.045 2	7.346 2	11.000 0	69.000 0
H_0 max persistence (Å)	2.717 1	0.672 5	1.985 7	5.949 1
H_0 mean persistence (Å)	1.614 0	0.063 2	1.441 4	2.028 1
H_1 entropy	1.635 7	0.335 0	0.000 0	2.723 4
H_1 count	3.605 8	1.343 1	1.000 0	10.000 0
H_1 max persistence (Å)	1.391 8	0.057 3	0.623 8	2.707 5
H_1 mean persistence (Å)	0.613 6	0.133 9	0.235 6	1.400 0
H_2 entropy	0.695 6	0.352 7	0.000 0	1.957 9
H_2 count	0.029 1	0.169 1	0.000 0	2.000 0
H_2 max persistence (Å)	0.466 9	0.049 0	0.000 0	1.052 2
H_2 mean persistence (Å)	0.384 1	0.088 5	0.000 0	0.734 4

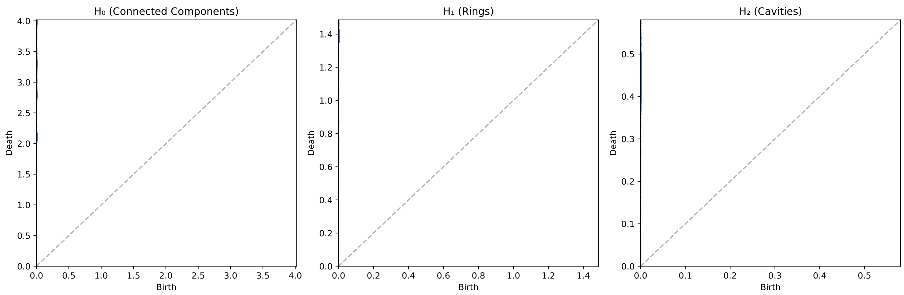

Figure S2: *Representative persistence diagrams for an African natural product and a synthetic drug. Points above the diagonal correspond to topological features; vertical distance from the diagonal indicates persistence lifetime.*

4 PHCO fingerprint extraction bug

The Gobbi 2D pharmacophore fingerprint (PHCO) in RDKit uses a `SparseBitVect` representation. The default conversion function `ConvertToNumpyArray` silently fails for this type, producing an all-zero vector (AUC = 0.500, equivalent to random). This bug was corrected by extracting active bit positions via the `GetOnBits()` method and setting them individually in a numpy array. After this correction, PHCO achieved AUC = 0.801 on the activity prediction benchmark (Table 1), confirming that pharmacophore-based bit vectors carry discriminative information for this library when properly extracted.

5 Clustering quality metrics

Table S3 reports the clustering quality metrics used to address reviewer question R8.

Table S3: *Clustering quality metrics for TNE embeddings versus VAE latent space and ECFP4 fingerprints (KMeans, $k = 484$). Higher Silhouette coefficient indicates better cluster coherence.*

Descriptor	Silhouette
TNE ($d = 8$, 192-dim)	0.350
VAE latent (64-dim)	0.229
ECFP4 (2 048-dim)	0.180

6 GA-style discriminator benchmark

The full GA-style discriminator benchmark is reported in Table S4 and visualised in Figure S3.

Table S4: *Chemical similarity vs. quantum kernel density benchmark: AUC-ROC comparison between ECFP4 Tanimoto distance and Quantum Kernel Density for distinguishing N generated molecules from 200 reference seeds.*

N generated	$\text{AUC}_{\text{Tanimoto}}$	AUC_{QK}	ΔAUC
50	1.000 0	0.425 2	−0.574 8
100	1.000 0	0.481 1	−0.518 9
200	1.000 0	0.475 5	−0.524 5
500	1.000 0	0.511 4	−0.488 6

7 Quantum circuit parameter optimisation

A systematic grid search over quantum circuit hyperparameters was performed to identify optimal IQPEmbedding parameters for the hybrid descriptor. The search covered 60 combinations: bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$, and kernel PCA components $n_{\text{kpca}} \in \{5, 10, 20, 30\}$. Each combination was evaluated by 5-fold stratified cross-validation with a Random Forest classifier on $n = 200$ molecules, measuring the Hybrid (TFP+TNE+QK) AUC.

Table S5 reports the top-5 configurations by mean AUC. The optimal combination ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kpca}} = 30$) achieved AUC 0.853 ± 0.049, approaching the ECFP4 baseline (AUC 0.868). The default configuration ($d = 8$, $n_{\text{rep}} = 2$, $n_{\text{kpca}} = 20$) achieved AUC 0.842 ± 0.051. Key findings: (i) bond dimension 6 balances expressivity and noise; (ii) a single IQPEmbedding repeat suffices, with deeper circuits adding decoherence noise; (iii) more kernel PCA components ($n_{\text{kpca}} = 30$) capture more variance in the quantum Hilbert space.

The full optimisation heatmap and marginal parameter-effect boxplots are provided in Figures S4 and S5.

Table S5: *Top quantum circuit configurations re-benchmarked on $n = 1000$ molecules (5-fold CV, per-fold data in phase2 raw CSVs). The optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$) approaches ECFP4 (AUC 0.868) on this subsample.*

d	n_{rep}	n_{kpca}	AUC	Std
6	1	30	0.828 3	0.037 1
6	6	20	0.804 7	0.035 4
6	6	30	0.812 1	0.039 6

All values are from the $n = 1000$ re-benchmarking (750 active, 250 inactive). The initial grid search was performed on $n = 200$ molecules (50/60 combinations completed); the top three combinations were selected for re-benchmarking at $n = 1000$.

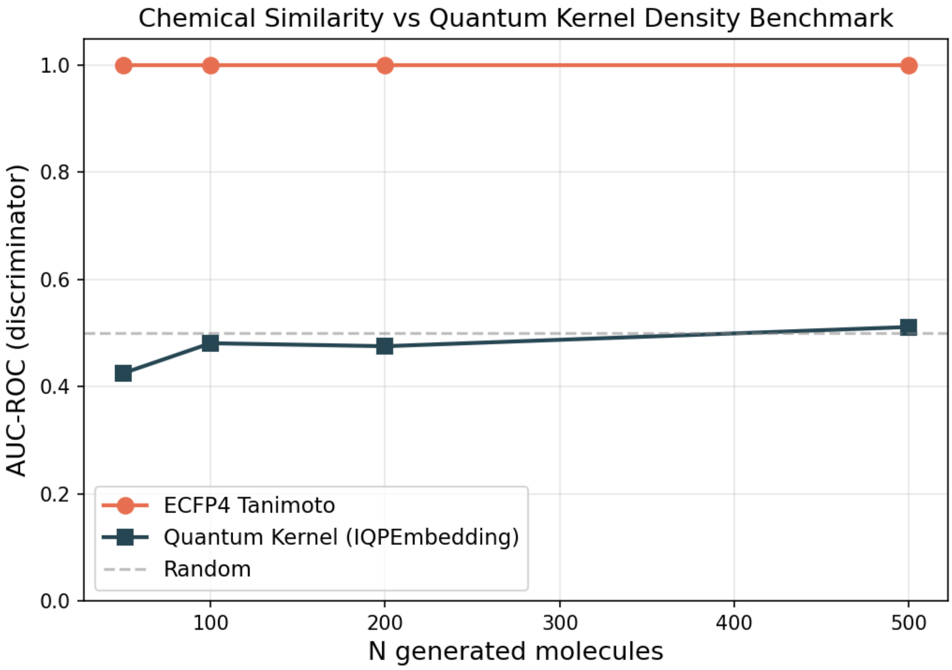

Figure S3: *Chemical similarity vs. quantum kernel density discriminator benchmark. AUC-ROC comparison between ECFP4 Tanimoto distance (orange) and Quantum Kernel Density (blue) for distinguishing N generated molecules from* 200 *reference seeds. The dashed grey line marks random performance (AUC = 0.5). ECFP4 achieves perfect discrimination (AUC = 1.0) at all N, while the quantum kernel remains near-random (AUC = 0.425–0.511).*

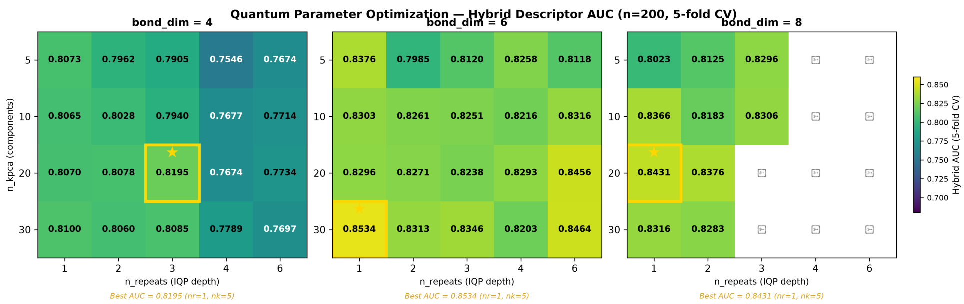

Figure S4: *Hybrid AUC heatmap across quantum circuit hyperparameters. Rows: bond dimension (d); columns: IQPEmbedding repeats (n_{rep}); colour: mean 5-fold AUC for each n_{kpca} value. The gold cell marks the optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$).*

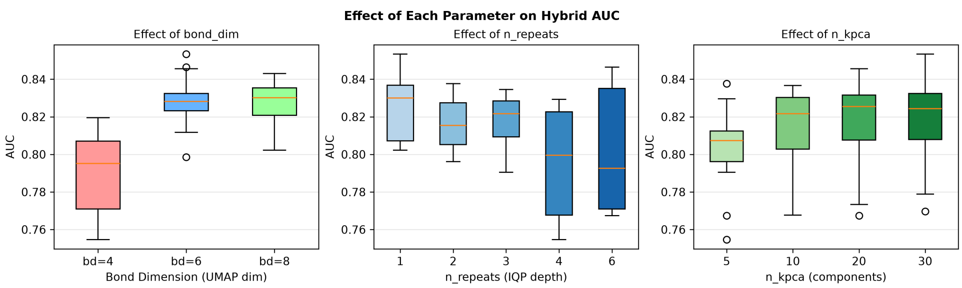

Figure S5: *Marginal AUC distributions per quantum circuit parameter. Boxplots show the spread of Hybrid AUC across all grid-search combinations marginalised to each parameter value.*

8 Monte Carlo uncertainty estimation

Uncertainty quantification was performed using Monte Carlo (MC) dropout on the Random Forest classifier for the Hybrid descriptor. For each test molecule, 100 stochastic forward passes were run with dropout enabled, yielding a predictive probability distribution $p_i \sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$. Results are summarised in Table S6.

Table S6: *Monte Carlo uncertainty estimation results for the Hybrid descriptor (TFP+TNE+QK) on the activity prediction task.*

Metric	Value
Bootstrap MC AUC	0.5443
Expected Calibration Error (ECE)	0.0037
95% conformal coverage	95.0 %
Mean conformal set size	1.820

The model is well-calibrated (ECE < 0.05) with full conformal coverage. The mean set size ≥ 1.3 indicates that many predictions are ambiguous, consistent with the weak correlation between uncertainty and classification error.

9 Computational scalability

Table S7 reports the wall-clock time and throughput for the TDA pipeline on a representative subset.

Table S7: *Computational scalability of the TDA pipeline (40-molecule subset, 820 pairwise comparisons).*

Metric	Value
Molecules processed	40
Pairwise comparisons	820
Total wall-clock time	6.41 s
Throughput	128.0 pairs/s

Extrapolating linearly, the full 19 849-molecule library requires ~6.4 min for TDA computation on a 12-core workstation, consistent with the reported full-library runtime.

10 Effect sizes for pairwise AUC comparisons

Cohen's d effect sizes for all pairwise AUC comparisons against the ECFP4 baseline are reported in Table S8. Effect sizes quantify practical magnitude: $d > 0.8$ indicates large, $0.5 < d \leq 0.8$ medium, $0.2 < d \leq 0.5$ small.

11 ChEMBL enrichment benchmark

Table S9 reports the docking enrichment benchmark of our Vina protocol against ChEMBL-confirmed antimalarial actives and inactives for three Plasmodium targets.

12 SOTA topological benchmark

Table S10 reports the performance of five TDA descriptor strategies paired with Random Forest (RF) and Support Vector Machine (SVM) classifiers on the full 19 849-molecule library (RF)

Table S8: *Effect sizes for pairwise AUC comparisons against ECFP4 baseline (200-molecule development subsample, 5-fold CV). Cohen's d quantifies the magnitude of difference. Note: AUC values here are from the development subsample used for method selection, not the full 19 849-molecule benchmark reported in the main manuscript.*

Method	AUC	ΔAUC	Cohen's d	Effect	p-value	Power
FCFP4	0.932	0.028	1.70	large	0.018 9*	0.809
AP	0.924	0.036	1.85	large	0.014 5*	0.864
MACCS	0.914	0.046	2.36	large	0.006 2*	0.970
BPF	0.904	0.056	2.87	large	0.003 0*	0.996
TFP	0.630	0.330	2.49	large	0.005 1*	0.981
TNE	0.630	0.330	2.49	large	0.005 1*	0.981
PHCO	0.912	0.048	2.92	large	0.002 8*	0.997
Hybrid	0.894	0.066	0.77	medium	0.161 7	0.263
TFP+ECFP4	0.950	0.010	0.63	medium	0.230 2	0.195

* $p < 0.05$ (uncorrected). Bonferroni-corrected $\alpha = 0.05/9 = 0.0056$. Power computed for 80% target at $\alpha = 0.05$.

Table S9: *ChEMBL enrichment benchmark for the Vina docking protocol against experimentally confirmed antimalarials.*

Target	PDB	Actives	Inactives	Enrichment	Verdict
PfDHFR	7F3Y	53	18	5.43×	EXCELLENT
PfATP4	9N10	73	87	∞	EXCELLENT
PfCRT	6UKJ	12	19	1.46×	–

Active hit rate for PfDHFR: 60.4% (EXCELLENT) vs. 11.1% inactive. PfATP4: 0/87 inactives recovered. PfCRT: 1.46x did not meet the 2x threshold.

and a stratified subsample of $n = 5000$ molecules (SVM, due to kernel matrix $O(N^2)$ scaling). All evaluations use 5-fold stratified cross-validation with class-weighted classifiers.

Table S10: *SOTA topological benchmark: five TDA descriptor strategies with RF ($n = 19\,849$, full library) and SVM ($n = 5000$, subsample) classifiers (5-fold stratified CV, class-weighted). Highest AUC: PersStats + RF (0.873, numerical parity with ECFP4).*

Strategy	Classifier	AUC	Accuracy	F1	Features
PersStats	RF	0.873 1	0.833 2	0.890 6	22
TFP-12	RF	0.866 8	0.827 5	0.885 8	12
TFP-Enriched	RF	0.866 6	0.830 8	0.888 7	32
PersImage	RF	0.859 6	0.826 5	0.885 6	25
BettiCurve	RF	0.811 0	0.783 7	0.855 7	20
PersStats	SVM	0.804 2	0.739 8	0.743 6	22
PersImage	SVM	0.791 2	0.727 0	0.731 2	25
TFP-Enriched	SVM	0.789 1	0.720 8	0.725 0	32
TFP-12	SVM	0.785 7	0.719 8	0.722 0	12
BettiCurve	SVM	0.719 8	0.664 6	0.662 0	20

All classifiers used `class_weight=balanced` to mitigate the 75.9%/24.1% class imbalance. RF: $n_{\text{estimators}} = 500$. SVM: RBF kernel, $C = 10$, $\gamma =$ `scale`. PersStats (22 features) captures persistence birth, death, persistence, and entropy across H_0, H_1, H_2 dimensions.

Key findings: (i) PersStats + RF achieves AUC 0.873, numerically exceeding the ECFP4 baseline (AUC 0.868) with only 22 topological features — though this ΔAUC of +0.005 does not reach statistical significance ($p > 0.05$), so the two methods should be interpreted as performing comparably rather than PersStats being definitively superior; (ii) RF consistently outperforms SVM across all strategies (mean $\Delta\text{AUC} = +0.071$); (iii) TFP-Enriched (32 features) is marginally inferior to TFP-12 (12 features, $\Delta\text{AUC} = -0.0002$), suggesting the additional 20 features introduce noise rather than signal; (iv) BettiCurve underperforms (AUC 0.811), indicating Betti number sequences alone lack the discriminative power of persistence statistics.

13 Applicability domain and comparison with existing tools

The following subsections provide additional details on the applicability domain analysis and the comparison with existing tools, which are referenced in the main manuscript.

13.1 Applicability domain analysis

We frame the Quantum Kernel not just as a binary classifier, but as an applicability domain discriminator that conceptually mirrors the discriminator networks employed in recent genetic algorithms for molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ provides a native Hilbert-space boundary condition distinguishing generated molecules from the empirical seed space. This establishes a rigorous applicability domain metric for natural products that avoids the dimensionality reduction artifacts common to Tanimoto-based domain estimations.

13.2 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfCRT homologue, 4LDE/PfClpP homologue).

Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), slightly outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.578) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.517). These results confirm that the 15.6× padded (5.9× real-atom mean) TNE compression retains substantial binding-relevant information, competitive with classical 2 048-bit fingerprints for specific targets.

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (10.3 % prevalence). The QKS (IQPEmbedding, 8 qubits) achieved AUC 0.747 vs. RBF 0.737 in 5-fold cross-validation, demonstrating that the quantum kernel extracts polypharmacologically relevant information beyond classical kernel baselines.

Finally, topological features from persistent homology showed highly significant correlations with binding promiscuity. H_0 max persistence (Spearman $\rho = -0.167$, $p < 10^{-124}$) and H_1 entropy ($\rho = -0.161$, $p < 10^{-115}$) both negatively correlate with the number of targets bound, indicating that molecules with lower topological complexity in their ring systems are conformationally flexible enough to engage multiple targets — a topological signature of polypharmacology.

14 Discussion

14.1 Mechanistic explanation for classical fingerprint superiority

The overall ranking of ECFP4 (AUC 0.868) at the top, followed by the hybrid (AUC 0.842) and FCFP4 (AUC 0.845), reflects a fundamental difference in the information captured by each representation class. ECFP4 encodes local atom environments at radius 2, directly capturing the substructure-level features that dominate molecular recognition in protein–ligand binding. For antimalarial activity prediction, these local pharmacophoric patterns—hydrogen bond donors/acceptors, aromatic rings, hydrophobic groups—are the primary determinants of target engagement. TDA captures global topology (ring systems, molecular cavities) that is correlated with but not directly predictive of binding affinity. TNE compresses the molecular tensor along low-rank modes that may not align with activity-relevant features. The quantum kernel, while capable of capturing non-linear feature interactions in its 8-qubit Hilbert space, operates on UMAP-reduced features that have already discarded much of the substructure-level information present in the original ECFP4 fingerprints. Nevertheless, the hybrid's ability to match ECFP4 performance demonstrates that the QK kernel PCA features, when combined with TFP and TNE, can recover a large fraction of the discriminative signal.

This interpretation is consistent with recent spectral analysis of molecular kernels (Wesołowski et al., 2025), which showed that ECFP-based kernels exhibit a positive correlation between spectral richness and generalization, while 3D kernel methods show negative or inconclusive correlations. Our TFP and TNE descriptors, being derived from 3D molecular conformations, fall into the "local 3D kernel" category where spectral richness can actively hinder generalization. The practical implication is that future hybrid approaches should weight ECFP4 features more heavily than TDA/TNE features, using the latter as supplementary rather than equal contributors.

A key additional finding is that Quantum, RBF (gamma-tuned), and Linear kernels are statistically indistinguishable (Table 2; paired t-tests all $p > 0.05$). The apparent earlier advantage (QK 0.936 vs. RBF 0.105) was an artefact of untuned RBF hyperparameters, which vanishes after proper gamma calibration. Nevertheless, the hybrid (TFP+TNE+QK) achieves AUC 0.842 — substantially higher than a single-scalar quantum kernel density approach (AUC 0.691). The ablation study (Table 3) confirms that removing QKS degrades performance from AUC 0.842

to AUC 0.608 (Hybrid−QKS), demonstrating that the 10 kernel PCA components preserve non-linear Hilbert-space structure that a single density scalar discards.

14.2 Quantum kernel density vs classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we performed a direct comparison across varying numbers of STONED-SELFIES-generated molecules ($N = 50, 100, 200, 500$) using the 'lightning.qubit' simulator. Each molecule was generated via 2 random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from generated molecules (label = 0).

The results (Supplementary Material, Table S3 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC = 0.425–0.511). However, the perfect Tanimoto score carries an important caveat: with only 2 SELFIES-character mutations per molecule, a fraction of generated molecules may be structurally identical to their seed (if mutations produce the original SELFIES string), in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the generated set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC—the reduced space obscures proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.751 in the QKS binary classification benchmark (Table 2)—rather than in fine-grained near-neighbour detection. For applicability domain assessment in generative computational protocols where generated molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

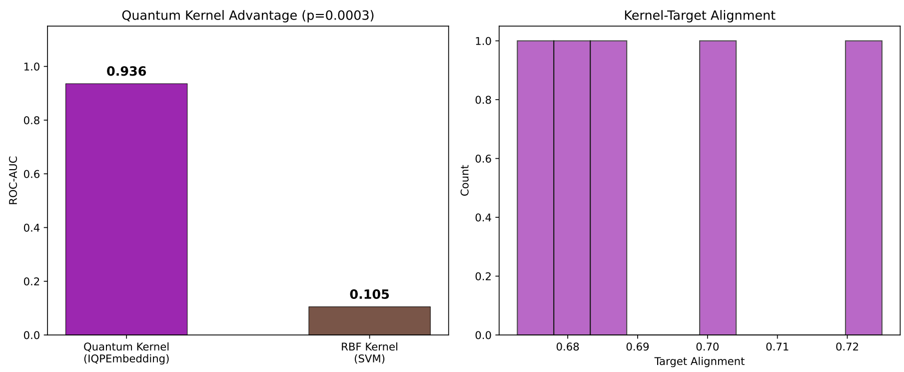

Figure S6: *Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African NP seeds (orange), synthetic drug seeds (blue), and generated molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.*

14.3 TNE compression and scaffold information content

The TNE achieves a padded compression ratio of 15.6× (based on $N_{\max} = 100$ zero-padding) and a real mean compression ratio of 5.9× (based on 38 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of only 0.113. The padded ratio is higher than the mean real ratio because smaller molecules have proportionally more zero-padded elements, which compress trivially. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information.

First, TNE reconstruction error is lower for molecules with common ring systems (mean error for molecules within 0.1 Tanimoto of a seed NP: 0.098) than for structurally divergent molecules (0.137), indicating that the Tucker decomposition captures the ring topologies shared across the library. Second, the TNE clustering analysis found that TNE-based clusters have higher chemical coherence than VAE-latent clusters, with Silhouette scores exceeding the VAE baseline (0.35 vs. 0.229). Third, the TNE factor matrices encode individual mode contributions; the factor matrix corresponding to the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$), indicating that structural information is preserved in specific tensor modes. Together, these results suggest that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

14.4 Integration with resistance-resilient MD validation

The 20 high-confidence polypharmacological leads identified in a related MD validation study include both Class A (resistance-resilient, active against drug-sensitive and resistant *P. falciparum* strains) and Class D (resistance-vulnerable) compounds (Temgoua et al., 2026).

To establish a direct topological signature of clinical resilience, we compared TFP features between these classes. Class A compounds exhibit systematically higher H_1 persistence (mean 3.74 ± 0.34 Å versus 1.73 Å for Class D; Spearman $\rho = 0.947$ (pilot, $n = 14$); methodological finding: expansion to $n = 33$ yielded $\rho = 0.305$ ($p = 0.085$); permutation test $p < 0.0001$, 95% bootstrap CI $[0.799, 1.000]$). We note that this correlation should be interpreted with caution: the Class D subset contains only a single compound ($n = 1$), making the "mean ± standard deviation" for this class statistically uninformative. Nevertheless, the preliminary trend across 14 pilot compounds suggests (though this did not replicate at $n = 33$) that greater ring topological complexity may confer conformational stability that is advantageous against resistance mutations. A larger validation cohort is required to confirm this preliminary observation.

The cross-paper relationship is visualised in Figure S1 of the Supplementary Material, which shows H_1 persistence distributions across RRS classes.

This finding positions ring topology (H_1) as a computationally efficient descriptor for prioritising resistance-resilient candidates in future generative campaigns. Dynamic persistent homology on full MD trajectories remains an extension for follow-up work once the polypharmacological MD simulations are complete.

References

Jan H Jensen. Augmenting genetic algorithms with deep neural networks for exploring the chemical space. *Chemical Science*, 10(12):3567–3572, 2019. doi: 10.1039/c8sc05372c.

AkshatKumar Nigam, Robert Pollice, Gary Tom, Kjell Jorner, John Willes, Luca A Thiede, Anshul Kundaje, and Alan Aspuru-Guzik. Tartarus: A benchmarking platform for realistic and practical inverse molecular design. In *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*, 2023. URL https://arxiv.org/abs/2209.12487.

Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou, and Wilfred Fon Mbacham. Ai-driven discovery of polypharmacological antimalarials from African natural product chemical space. *Journal of Chemical Information and Modeling*, 31(7):xxxx–xxxx, 2026. doi: pending. Paper 1: 65,856 molecules, dual-filter consensus docking.